

PRACTICAL WEEK – 7

TASK 1: Write a program that prints the addresses Of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment

```
keerthi_priya@LAPTOP-OSGV × + v
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void code_function()
{
    printf("This is a code function.\n");
}

int main()
{
    int stack_var = 30;
    static int local_static_var = 40;

    int *heap_var = (int *)malloc(sizeof(int));
    *heap_var = 50;

    printf("Process ID: %d\n\n", getpid());

    printf("Code address      : %p\n", (void *)code_function);
    printf("Global variable   : %p\n", (void *)&global_var);
    printf("Static variable      : %p\n", (void *)&static_var);
    printf("Local static variable : %p\n", (void *)&local_static_var);
    printf("Heap variable        : %p\n", (void *)heap_var);
    printf("Stack variable       : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");
    printf("Press Ctrl+C to stop.\n");

    while (1)
    {
        sleep(2);
    }

    free(heap_var);

    return 0;
}
```

```
keerthi_priya@LAPTOP-OSGV x + v
keerthi_priya@LAPTOP-OSGVCE7A:~$ mkdir practical_memory
keerthi_priya@LAPTOP-OSGVCE7A:~$ cd practical_memory
keerthi_priya@LAPTOP-OSGVCE7A:~/practical_memory$ nano memory_layout.c
keerthi_priya@LAPTOP-OSGVCE7A:~/practical_memory$ gcc memory_layout.c -o memory_layout
keerthi_priya@LAPTOP-OSGVCE7A:~/practical_memory$ ./memory_layout
Process ID: 533

Code address      : 0x5cf2d317d1c9
Global variable   : 0x5cf2d3180010
Static variable   : 0x5cf2d3180014
Local static variable : 0x5cf2d3180018
Heap variable     : 0x5cf2d91c6010
Stack variable    : 0x7ffe22b311dc

Process is running...
Press Ctrl+C to stop.
^C
keerthi_priya@LAPTOP-OSGVCE7A:~/practical_memory$ |
```

TASK 2: Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
keerthi_priya@LAPTOP-OSGV x + v
keerthi_priya@LAPTOP-OSGVCE7A:~$ nano memory_analysis.c
keerthi_priya@LAPTOP-OSGVCE7A:~$ ^C
-bash: :s^C: substitution failed
keerthi_priya@LAPTOP-OSGVCE7A:~$ gcc memory_analysis.c -o memory_analysis
keerthi_priya@LAPTOP-OSGVCE7A:~$ ./memory_analysis &
[1] 815
Process ID: 815
Code address      : 0x580edf6811c9
Global variable   : 0x580edf684010
Static variable   : 0x580edf684014
Local static variable : 0x580edf684018
Heap variable     : 0x580efb1e3010
Stack variable    : 0x7ffca246dc4c

Process is running...
keerthi_priya@LAPTOP-OSGVCE7A:~$ cat /proc/815/maps
580edf680000-580edf681000 r--p 00000000 08:30 32066 /home/keerthi_priya/memory_analysis
580edf681000-580edf682000 r-xp 00001000 08:30 32066 /home/keerthi_priya/memory_analysis
580edf682000-580edf683000 r--p 00002000 08:30 32066 /home/keerthi_priya/memory_analysis
580edf683000-580edf684000 r--p 00003000 08:30 32066 /home/keerthi_priya/memory_analysis
580edf684000-580edf685000 rw-p 00004000 08:30 32066 /home/keerthi_priya/memory_analysis
580efb1e3000-580efb204000 rw-p 00000000 00:00 0 [heap]
7dd4e1200000-7dd4e1228000 r--p 00000000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7dd4e1228000-7dd4e13c0000 r-xp 00028000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7dd4e13c0000-7dd4e140e000 r--p 001c0000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7dd4e140e000-7dd4e1412000 r--p 0020d000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7dd4e1412000-7dd4e1414000 rw-p 00211000 08:30 13647 /usr/lib/x86_64-linux-gnu/libc.so.6
7dd4e1414000-7dd4e1421000 rw-p 00000000 00:00 0
7dd4e14b6000-7dd4e14b9000 rw-p 00000000 00:00 0
7dd4e14be000-7dd4e14c0000 rw-p 00000000 00:00 0
7dd4e14c0000-7dd4e14c4000 r--p 00000000 00:00 0
7dd4e14c4000-7dd4e14c6000 r--p 00000000 00:00 0
7dd4e14c6000-7dd4e14c8000 r-xp 00000000 00:00 0
7dd4e14c8000-7dd4e14c9000 r--p 00000000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7dd4e14c9000-7dd4e14f8000 r-xp 00001000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7dd4e14f8000-7dd4e1503000 r--p 00030000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7dd4e1503000-7dd4e1505000 r--p 0003b000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7dd4e1505000-7dd4e1506000 rw-p 0003d000 08:30 13644 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7dd4e1506000-7dd4e1507000 rw-p 00000000 00:00 0
7ffca244e000-7ffca246f000 rw-p 00000000 00:00 0 [stack]
keerthi_priya@LAPTOP-OSGVCE7A:~$ |
```

GNU nano 8.7.1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void display()
{
    printf("Inside code segment\n");
}

int main()
{
    int stack_var = 30;
    static int local_static = 40;

    int *heap_var = malloc(sizeof(int));
    *heap_var = 50;

    printf("Process ID: %d\n", getpid());

    printf("Code address      : %p\n", (void *)display);
    printf("Global variable    : %p\n", (void *)&global_var);
    printf("Static variable       : %p\n", (void *)&static_var);
    printf("Local static variable : %p\n", (void *)&local_static);
    printf("Heap variable         : %p\n", (void *)heap_var);
    printf("Stack variable        : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");

    while (1)
    {
        sleep(2);
    }

    free(heap_var);

    return 0;
}
```